POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Chemistry

Semester: Spring

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is buffer solution? Write the mechanism of buffer action of acidic buffer with suitable example. Calculate the pH of buffer solution containing 400 ml of 0.3 M acetic acid and 200 ml of 0.6 M sodium acetate (K_a of acetic acid is 1.8×10^{-5}) 7

   b) What do you mean by standard hydrogen electrode? Calculate the emf of the following cell 8

   $Zn(s)/Zn^{+2}(0.1M)//Cu^{+2}(1.75M)/Cu(s)$ at 25°C

   $E°_{Zn^{++}/Zn} = -0.76V$

   $E°_{Cu^{++}/Cu} = +0.34V$

   $R = 8.314 Jmol^{-1}K^{-1}$

   $F = 96500$ C

2. a) Give reasons for the followings 8

   i. $TiCl_3$ is coloured but $TiCl_4$ is colourless.

   ii. Transition elements show variable oxidation state.

   iii. Zn^{+2} salts are white.

   iv. Cu is not considered as transition element.

   b) Define ionization energy. Explain the factors affecting negativity. How electron affinity of chlorine is higher than Fluorine. 7

3. a) What are reaction intermediates? Explain the structure and stability of different types of carbocation. 8

   b) What is SN1 reaction? Give reaction mechanism and stereochemistry of SN1 reaction in given reaction 7

   $(CH_3)_3CX + NaOH(aq.) \rightarrow (CH_3)_3C\text{-}OH + NaX$

4. a) What is polymerization reaction? Write the different types of polymerization with suitable examples for each. 8

b) What is rubber? Write the process of vulcanization of rubber. 7

a) Write principle and important applications of mass spectroscopy in analytical field. Write the applications of TLC 8

b) What is explosive? Give the methods of preparation of TNT and its important uses. 7

a) What is ozone layer depletion? Write its adverse effect to living beings? Describe its photochemistry 8

b) Define water pollution. Discuss briefly about its sources, effects and control methods 7

Write short notes on: **(Any two)** 2×5

a) Corrosion, its types and prevention

b) Difference between E_1&E_2 reactions

c) Thermoplastics and Thermosetting Plastics